

Digital Twins

Predictive Digital Twin for Crack Prevention in an FBG-Instrumented
Wind Turbine Blade Sample

Why monitoring wind turbine blades matters

Context

Wind turbine blades are large composite structures exposed to cyclic mechanical loads.

Early crack detection helps :

- Reduce maintenance **cost**
- Avoid **catastrophic failure**
- Increase **lifetime** of the physical asset

Problem example



© Commune de Seneffe, wind turbine in Feluy

Presentation roadmap

Physical twin

Composite wind-turbine blade sample instrumented with FBG sensors.

Data acquisition

FBG wavelength shifts are recorded during three-point bending tests.

Digital twin

A CNN-based predictive model estimates crack probability from strain-related features.

The digital twin is the full pipeline: sensing, preprocessing, feature extraction, and crack-risk prediction.

Physical Twin: Composite Blade Sample

Embedding FBG sensors inside a composite

The physical twin is a **lab-scale composite blade** sample made of fiberglass and epoxy resin.

The optical fiber containing the **FBG sensors** is embedded between the composite layers.

The fiber is not placed on the **neutral axis**, where bending strain would be close to zero.

Fabrication



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FBG sensing principle

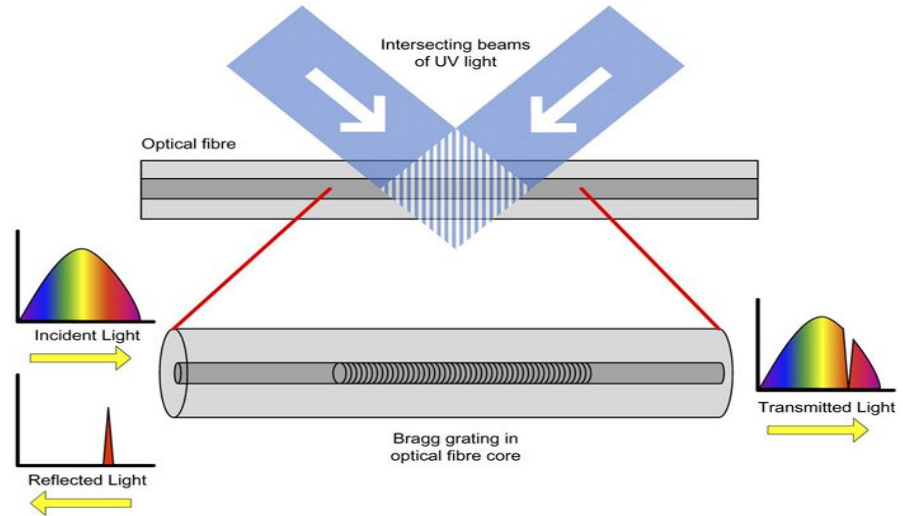
Principle

An FBG is a periodic modulation of the refractive index in the fiber core. It reflects one specific Bragg wavelength:

$$(1) \quad \lambda_{max} = 2 n_{core} \Lambda$$

When the structure is strained, Λ **change**, causing a measurable wavelength shift.

FBG sensing principle



© Spie News, <https://spie.org/news/2867-polymer-fiber-grating-sensors>

Three-point bending test

Loading the sample and recording FBG wavelength shifts

The composite sample is loaded in a **three-point bending** setup.

The **pressure** is increased step by step, then decreased to observe **repeatability** and possible **hysteresis**.

During the test, the interrogator records the **FBG wavelength shifts**.

Fabrication



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Data acquisition

Interrogator files

The interrogator records the Bragg wavelength peaks of the FBG sensors during the bending test.

The following data are recorded :

- Timestamp
- Time
- Peak wavelength (for each FBG)

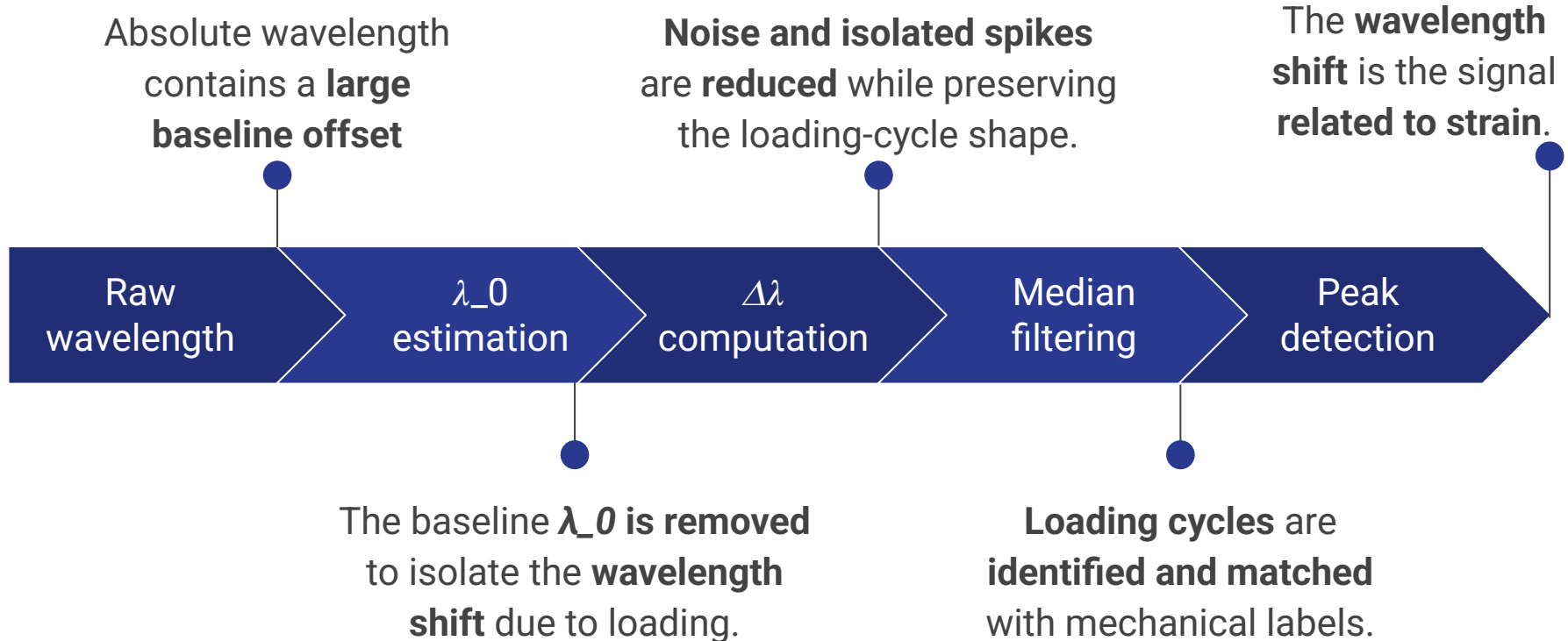
Hand-held Excel workbook

The interrogator records the Bragg wavelength peaks of the FBG sensors during the bending test.

Each sheet contains :

- Pressure
- Force
- Displacement
- Crack marker

From raw wavelength to clean signal



CNN input features

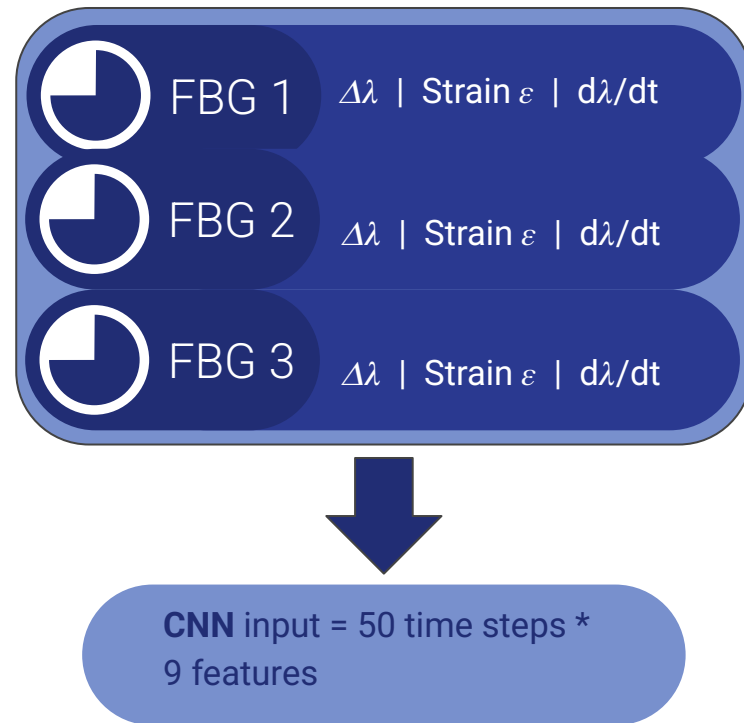
FBG signals into a fixed-size learning window

Each loading cycle is represented as a short time window.

For each FBG sensor, three features are used:

- Wavelength shift $\Delta\lambda$
- Estimated strain ϵ
- Wavelength derivative $d\lambda/dt$

With 3 FBG sensors, the model input is **50 time steps $\times$ 9 features**



CNN Crack Classifier

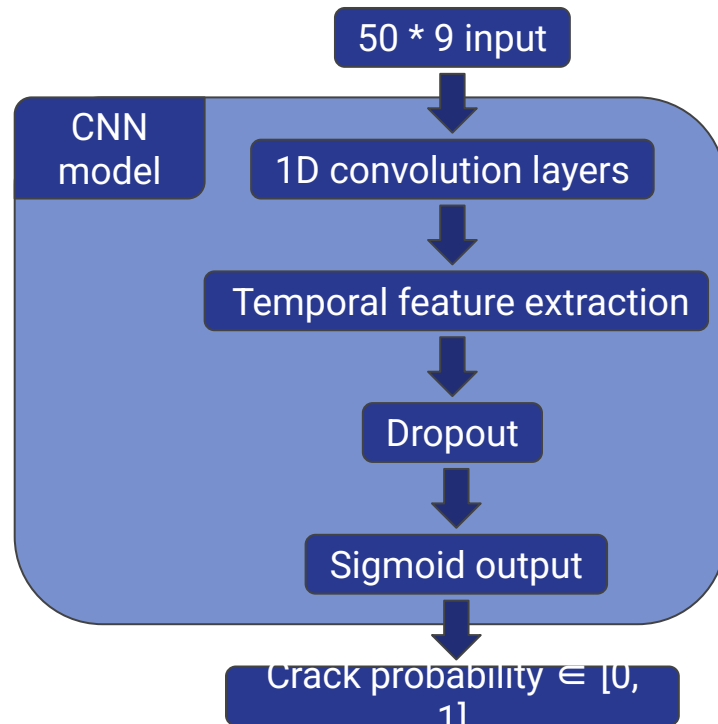
Patterns learning in the FBG strain signals

The **CNN** learns local temporal patterns in the FBG signals:

- Peaks
- Slope changes
- Abnormal strain evolution

Output:

A crack probability **between 0 and 1**



Physics-Informed Crack Prediction

Combining CNN features with Euler-Bernoulli theory

Euler-Bernoulli beam theory gives an expected strain during bending.

The model compares:

- Measured **strain from FBG sensors**
- Predicted **strain from force and geometry**

The residual helps detect **abnormal behaviour**.

Euler-Bernoulli beam theory

$$\varepsilon_{physics} = 3 F L y / (E b h^3)$$

<i>test set confusion matrix*</i>	predicted no crack	predicted crack
crack	0	8
no crack	86	14

**with crack-class weight = 26*

Conclusion

Achieved

- FBG-instrumented composite sample
- Preprocessing and feature extraction
- CNN-based crack probability prediction
- Physics-informed residual analysis

Limitations

- Small dataset
- Rare crack labels
- Approximate beam theory